This volume is for a reader who knows school algebra and has never been asked to write a proof. It teaches one skill: to read a mathematical statement exactly, and to prove it in sentences that a reader can check line by line.

It teaches logic as the grammar of proof. A proof is first written as a **derivation**, a chain in which every line is a premise, an assumption, or one application of a rule, so that checking it needs no idea: for each line we ask which rule produced it and from which lines. Each rule is introduced beside the sentence a mathematician writes when using it, and it is practiced at once, in short derivations and in proofs about numbers. The last chapter turns derivations back into ordinary mathematical prose, and shows that the rules are still there underneath.

What it covers.

Chapter	What it does
1	The format of a derivation, and the rules for "and", "implies" and "or"
2	The rules for "not" and "if and only if", proof by contradiction and the classical rules, and a method for finding derivations
3	Truth tables, induction on $\mathbb{N}$, and the theorems that the rules are sound and complete
4	"For every" and "there exists": terms and formulas, the quantifier rules and their side conditions, equality, and models
5	From derivations to mathematical prose: four levels of writing, habits for reading, writing and checking proofs, and a catalogue of the errors that recur

How to read it. Read the chapters in order: each uses the ones before it. Every section has Quick checks with folded answers; answer each one before you unfold it. The exercises of a section come in three groups, A to check understanding, B for practice and C to go deeper, and the last section of each chapter adds R items that reach back to earlier chapters. Do the derivations on paper, line by line, before reading the solution. A chapter's opening page lists what it needs, with links, and the time it takes. Second-reading folds hold edge cases and longer justifications; they can be left closed on a first reading, but they are not optional, since later proofs may use what they contain. Sections and results marked optional can be skipped, and nothing outside them depends on them.

Why rules before truth tables. Many logic books begin with truth tables, where the row "a false hypothesis makes $P \to Q$ true" has to be accepted as a convention. This book starts from the rules. Section 1.1 reads that row informally, as a promise that is not broken; Chapter 2 derives it from the rules, and Chapter 3 brings in the tables and proves that rules and tables agree.

What comes next. *Sets, Relations, and Cardinality* puts the logic to work on sets, functions, relations, induction and the sizes of infinite sets. Its third chapter ends with Gate 0, an examination on both volumes, and a reader who passes it is ready for *Analysis on the Real Line.*

Number Systems: From $\mathbb{N}$ *to* $\mathbb{R}$ builds the numbers themselves out of sets.